INCH-POUND

MIL-PRF-4541D (USAF)
2 June 1997
SUPERSEDING
MIL-L-4541C (USAF)
23 August 1983

PERFORMANCE SPECIFICATION
LUBRICATING AND SERVICING UNIT, AIR OPERATED
HIGH AND LOW PRESSURE DELIVERY

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Description. This document covers an air operated, lubricating and servicing unit for dispensing general purpose lubricants, gear and engine oils and chassis grease.

2. APPLICABLE DOCUMENTS

2.1 This section is not applicable to this specification.

3. REQUIREMENTS

3.1 Materials.

3.1.1 Metals. Metals used in the fabrication of the lubricating and servicing unit shall be of corrosion-resistant type or suitably treated to resist corrosion.

3.1.2 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be in intimate contact with each other.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: SA-ALC/TILDD, 485 Quentin Roosevelt Rd., Bldg 171, Kelly AFB, TX 78241-6524 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4930

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:37Z
Check the source to verify that this is the current version before use.

3.1.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2 Design.

3.2.1 Air Pressure. The lubricating and servicing unit shall function between 90 and 175 pounds per square inch (psi) and withstand stall pressure up to and including 200 psi air line pressure.

3.2.2 Maintenance. The lubricating and servicing unit shall be such that maintenance can be accomplished utilizing general purpose tools.

3.2.3 Performance. The lubricating and servicing unit shall meet the following performance requirements:

a. Operating temperatures ranging from -20° Fahrenheit (F) to 130°F and storage in temperatures ranging from -40°F to 160°F.

b. A relative humidity of 95 to 100 percent at an operating temperature of 130°F and a storage temperature of 160°F.

c. Lubricant displacements utilizing air pressures and ambient temperatures as indicated in Table I.

d. Operating as intended without leaking air, grease, or oil.

3.3 Cabinet. The cabinet shall be adequate to contain and provide easy access to three 120 pound drums, pumps, hose reels, hoses, grease and oil guns. Cabinet shall have a pulling-pushing handle, lock, and be mounted on a mobile base.

3.3.1 Base. The base shall withstand a minimum load of 800 pounds. The base shall be mounted on two wheels and two swivel casters. The base shall be equipped with a foot brake. The base shall also be equipped with four lifting and tie-down devices to facilitate loading, unloading, and servicing during transport.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:37Z
Check the source to verify that this is the current version before use.

3.4 Accessories. The following accessories shall be furnished with each lubricating and servicing unit.

Quantify	Item
1 each	Padlock with 2 keys.
1 each	Grease Gun, manually operated with a quick change feature.
1 each	Grease Gun, high pressure, with quick change feature.
1 each	Oil Gun, suction type.
1 each	Flexible extension.
1 each	Coupler, shafer type fitting.
1 each	Oiler, pump feed, with a rigid bent spout.
1 each	Oil spray Gun, air operated, with connector.
1 each	Adapter-Coupler, giant button-head, with 1/2 inch National Taper Pipe (NPT) female thread.

3.5 Pumps, lubricant.

3.5.1 Pumps, low pressure, gear and engine oils. Pumps shall be double acting, self priming, and air motor driven. Pumps shall be equipped with an air pressure gauge (0 to 200 psi scale), air pressure regulators and ball type foot valves. Performance: see Table I.

3.5.2 Pumps, high pressure, chassis grease. Pumps shall be double acting, self priming, and air motor driven. Pumps shall be equipped with an air pressure gauge (0 to 200 psi scale) and an air pressure regulator. Performance: see Table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:37Z
Check the source to verify that this is the current version before use.

TABLE 1. Lubricant Displacement

	MINIMUM DELIVERY PER MINUTE	AIR PRESSURE USED IN PSI	AMBIENT TEMPERATURE IN DEGREES FAHRENHEIT
High Pressure Pump Delivery (Using Chassis Grease)	20 Ounces	120	+70 ± 5
	14 Ounces	120	0 ± 2
	5 Ounces	120	-20 ± 2
NOTE: Static pressure developed by the pump shall be not less than 200 psi.			
Low Pressure Pump Delivery (Using Lubricating Oil, Gear) SAE 90 weight	16 Pounds	120	+70 ± 5
	12 Pounds	120	0 ± 2
	8 Pounds	120	-20 ± 2
Low Pressure Pump Delivery (Using Lubricating Oil, Internal Combustion)	3 Gallons	120	+70 ± 5
	1 3/4 Gallons	120	0 ± 2
	1 1/4 Gallons	120	-20 ± 2

3.5.3 Pump screen. The suction end of the gear and engine oil pumps and the chassis grease pump shall be fitted with a filtering device to prevent entrance of foreign matter into the pump. The filtering device shall be readily removable without disassembling any part of the pump for cleaning purposes.

3.6 Reel, chassis lubricant. A lubricant hose reel shall be provided and mounted within the cabinet. The reel shall be spring actuated with the capacity to accommodate 30 feet of 1/4 inch inside diameter (id) lubricant hose. The reel shall be provided with a latching or stop mechanism.

3.7 Hose.

3.7.1 High pressure. The hose shall be 1/4 inch id, minimum length of 25 feet with 1/4 inch NPT end fittings.

3.7.2 Low pressure. The hose shall be 3/8 or 1/2 inch id, minimum length of 13 feet with NPT end fittings.

3.8 Master valve assembly. A master air line valve with quick-disconnect coupler to relieve the air and lubricant pressure in the lines when unit is not in use shall be provided.

3.9 Manifold. An air manifold shall be provided to facilitate connection of air lines to the lubricant pumps. Manifold shall be equipped with quick-disconnect coupling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:37Z
Check the source to verify that this is the current version before use.

3.10 Control valve assemblies.

3.10.1 Control valve assembly, high pressure, chassis grease. A control valve assembly equipped with a swivel and an adapter, hydraulic coupler, shall be furnished. A pressure reducing device shall be provided. This reducing device shall function when at low pressure position, the chassis grease pump pressure ratio is reduced to approximately 1/8 the air pressure supplied to operate the pump, and when placed in the high pressure position the full pump pressure shall be available.

3.10.2 Control valve assembly, low-pressure, motor oil. A control valve equipped with a swivel shall be furnished for dispensing and metering motor oil. The totalizing meter shall measure in pints and quarts.

3.10.3 Control valve assembly, low-pressure, gear oil. A control valve equipped with a swivel shall be furnished for dispensing and metering gear oil. The totalizing meter shall measure in pints and quarts.

3.11 Air connection hoses. An air hose with one end terminating in a 1/4 inch external NPT end fitting and the opposite end terminating in a quick-disconnect shall be provided in the following locations:

a. From air manifold to gear oil pump.

b. From air manifold to motor oil pump.

c. From air manifold to chassis grease pump.

d. To connect air manifold, air line filter, and oiler, and master shut off valve.

3.12 Preparation, primer and paint. All painted surfaces shall be prepared and primed using procedures compatible with the final paint. The final paint shall be a standard, commercially available, non-absorbing gloss enamel.

3.13 Workmanship. The workmanship shall be high quality when compared to industry standards. There shall be no rough edges on machined or flame cut components. All welds shall be smooth and continuous. The welding procedures shall be conform to a welding code acceptable to the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:37Z
Check the source to verify that this is the current version before use.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall consist of all tests as described under para. 4.4.

4.3 Conformance inspection. Conformance inspection shall consist of all tests described under para. 4.4.1.

4.4 Examination and tests.

4.4.1 Examination of product. Each unit shall be examined to assure compliance with this specification relative to the requirements for materials, design, accessories, pumps, hoses, control valve assemblies, connection hoses, finish, primer and paints, proper assembly, completeness, marking and overall quality of workmanship as specified in para. 3.1, 3.3, 3.4, 3.6, 3.7, 3.8, 3.9, 3.10, 3.11, 3.12, and 3.13. The units shall be examined to assure conformance and documentation shall be provided to assure that the units meet the requirements of this specification.

4.4.2 Operation. One unit, representative of the production lot, shall be charged with lubricant and connected to a source of compressed air conforming to 3.2.1 and tested for proper operation of pumps, valves, et cetera. The unit shall be tested to verify compliance with 3.2 and 3.5. The pumps specified in para. 3.5.1 and 3.5.2 shall meet the performance requirements of Table 1. During these tests, a check shall be performed to determine if there are any air or lubricant leaks in the system. In addition, all components subject to internal either air, grease, or oil pressure shall be subjected to a hydrostatic test pressure of at least 200 pounds per square inch gauge (psig) for at least 5 minutes without either leaking, damage, or permanent deformation.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:37Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The lubricating and servicing unit covered by this specification is intended to be used for the lubrication of various types of automotive and heavy construction equipment.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Accessories to be furnished with the lubricating and servicing unit (see 3.4).

c. Technical publications to be furnished (see 3.13).

6.3 Subject term (key word) listing.

lubricating and servicing unit	chassis
pump low pressure	air operated
pump high pressure	high pressure
lubricant	low pressure

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 82

Agent:
Air Force - 99

(Project: 4930-0100)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:37Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:37Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. *The preparing activity must provide a reply within 30 days from receipt of the form.*

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-4541D	2. DOCUMENT DATE (YYMMDD) 970602

3. DOCUMENT TITLE LUBRICATING AND SERVICING UNIT, AIR OPERATED HIGH AND LOW PRESSURE

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	7.DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME RONETTE FELTER	b. TELEPHONE *Include Area Code)* (1) Commercial (512) 925-6314	(2) AUTOVON 945-6314
c. ADDRESS *(Include Zip Code)* SA-ALC/TILDD 485 QUENTIN ROOSEVELT RD., BLDG 171 KELLY AFB, TX 78241-6524	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026-01-17T03:37Z
Check the source to verify that this is the current version before use.